DIGITAL SIGNAL ENCODER AND DECODER

ITT 036 – Programming Assignment 1

Autumn Semester 2026

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1. Project Team

S. No.	Student Name	Enrollment No.
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2. Objective

The objective of this project is to develop a digital signal encoder and decoder with a graphical interface. It demonstrates digital data generation, line coding, scrambling, and analog-to-digital conversion. The application also provides visualization of encoded waveforms.

3. Functional Specifications

- **Digital Data Generation:** Generate random binary sequences, insert sequences of four or eight consecutive zeros, and identify the longest palindromic substring.
- **Line Coding:** Implement NRZ-L, NRZ-I, Manchester, Differential Manchester, Alternate Mark Inversion (AMI), and Return-to-Zero (RZ).
- **Scrambling:** Implement B8ZS and HDB3 zero-substitution techniques using bipolar AMI conventions.
- **Analog-to-Digital Conversion:** Implement Pulse Code Modulation (PCM) and Delta Modulation (DM), including encoding and decoding of sample sequences.
- **Graphical Interface:** Provide a JavaFX interface for selecting encoding techniques, entering data, processing signals, and viewing waveform results.

4. Technical Specifications

- **Language:** Java 21
- **GUI Framework:** JavaFX 21
- **Build Tool:** Apache Maven
- **Testing:** JUnit 5
- **Design:** Strategy Pattern with a central line-coding registry and a common codec interface.

Each line-coding scheme is implemented as an independent strategy. The registry provides access to the available strategies, while the `SignalCodec` facade provides common encoding and decoding operations. Input validation is performed to handle invalid bit streams, signal values, and analog parameters.

5. Execution and Testing

From the project root directory, execute:

- `mvn clean package` – Build the project.
- `mvn test` – Execute automated tests.
- `mvn javafx:run` – Launch the application.

The latest reported test run completed with **56 tests passed, 0 failures, 0 errors, and 0 skipped tests**.

6. Scope and Limitations

The application decodes signals using their numerical encoded values. Image-based waveform decoding is not implemented. The analog conversion techniques covered are PCM and Delta Modulation.

7. Acknowledgment

This project was developed for academic purposes as part of ITT 036. External resources and AI assistance, where used, should be acknowledged according to the course requirements.